United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Exam, Fall 2025

CSE 3811 : Artificial Intelligence

Total Marks: **30** Duration: **1 hour 30 Minutes**

Answer all the questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

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1. A city hospital uses an intelligent robot to deliver medicines from the pharmacy to patient rooms through corridors and intersections. Some corridors may be blocked, and the robot has a limited battery. The robot starts at the pharmacy, must deliver medicine to a specific patient room, and should avoid blocked corridors while minimizing battery usage. Each movement from one location to another consumes one unit of battery.
   a) Write the **PEAS** description for the robot. [2]
   b) Formulate this problem as a search problem. [4]

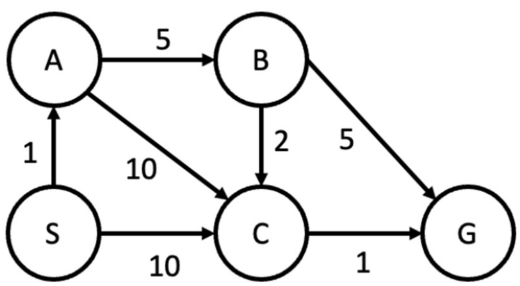

Figure 2(a)

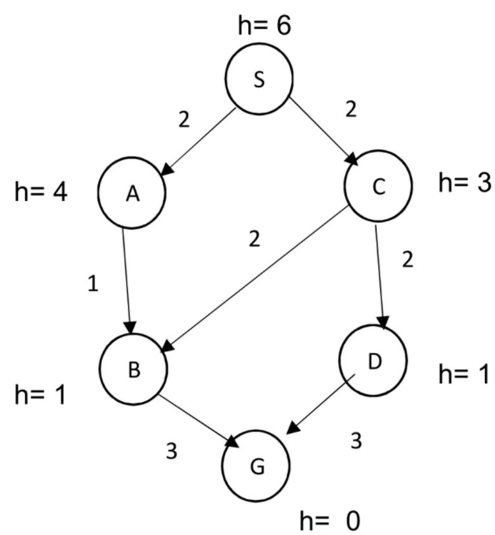

Figure 3(a)

2. a) Show the corresponding tree representation of the following state space graph and apply the following search strategies showing the order of expansion, solution paths and path costs (where S is the start state and G is the goal state). [2+2]
   i) **DFS**
   ii) **UCS**

   b) Suppose you are trying to solve a search problem with a search tree having **branching factor b**, **maximum depth m**, and **shallowest solution depth s**. The root is at level 0 and the search algorithms operate **from left to right**. Additionally, assume that expanding any node has **unit cost,** the search stops immediately upon finding the **first goal.** [1+1]

i) Which search strategy is guaranteed to expand **the fewest nodes for the following two cases**?

a) When the shallowest solution is at the leftmost node

b) When the shallowest solution is at the rightmost node

ii) How would your answers change if the search tree had **infinite depth** ($m = \infty$) and solutions are guaranteed to exist?

3. Consider the graph at figure 3(a) as state space graph where S is the start state and G is the goal state. h represents the heuristic values of the states.

a) Show the solution paths and the associated actual path costs for the following search algorithms: [2+2]

i. **A* Tree Search**

ii. **A* Graph Search**

b) Will A* graph search provide an optimal solution for the given state-space graph? If not, what is the particular reason behind it, explain briefly. [2]

4.

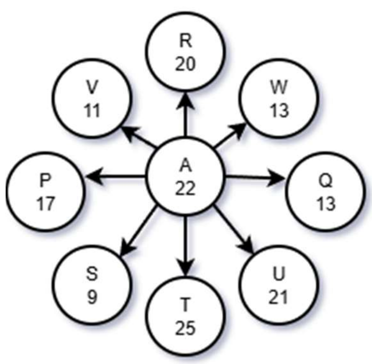

Figure 4(a)

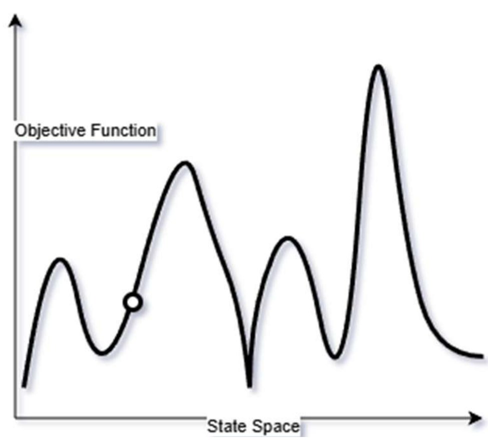

Figure 4(b)

a) Santa Claus is currently at node A and wants to travel to a new node. He decides to use the simulated-annealing algorithm to pick his next node. He used an objective function to get the values for each node. [4]

The Temperature(T) at any time-step(t) is given by:

$$T = t - 2t^2 + 67$$

The Random Generator generates any number(N) between 0 to 1 by following the function:

$$N = |cos(185 + 2.7t)|$$

Here cosine is in degrees and t is the time-step.

Starting from timestep t=0, one neighbor will be selected for calculation for each timestep (including t=0) according to the alphabetical order.

According to the algorithm, which node will he decide to **travel to first** and at what **timestep** will he make that decision? Show your calculations.

b) Determine which algorithm between stochastic-hill-climbing and simulated-annealing has the ability to give the best answer or if they are both equally good for the mentioned state-space graph with the given initial state. Include your reasoning. [2]

5. Consider the following game tree. The children of each node must be visited from left to right.

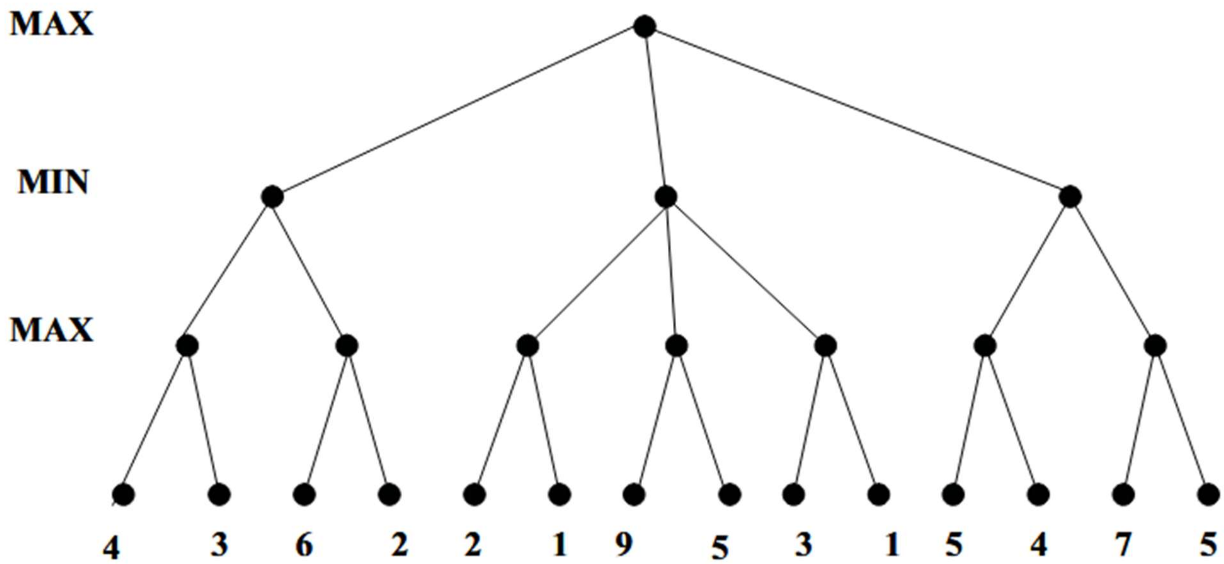

Figure 5(a)

a) Apply the MinMax algorithm with alpha-beta pruning to show which nodes will be pruned. Show utility[α , β , v] values for each node. [left to right] [4]

b) Does MinMax algorithm always provide optimal solution? If not give an example of such case. [2]